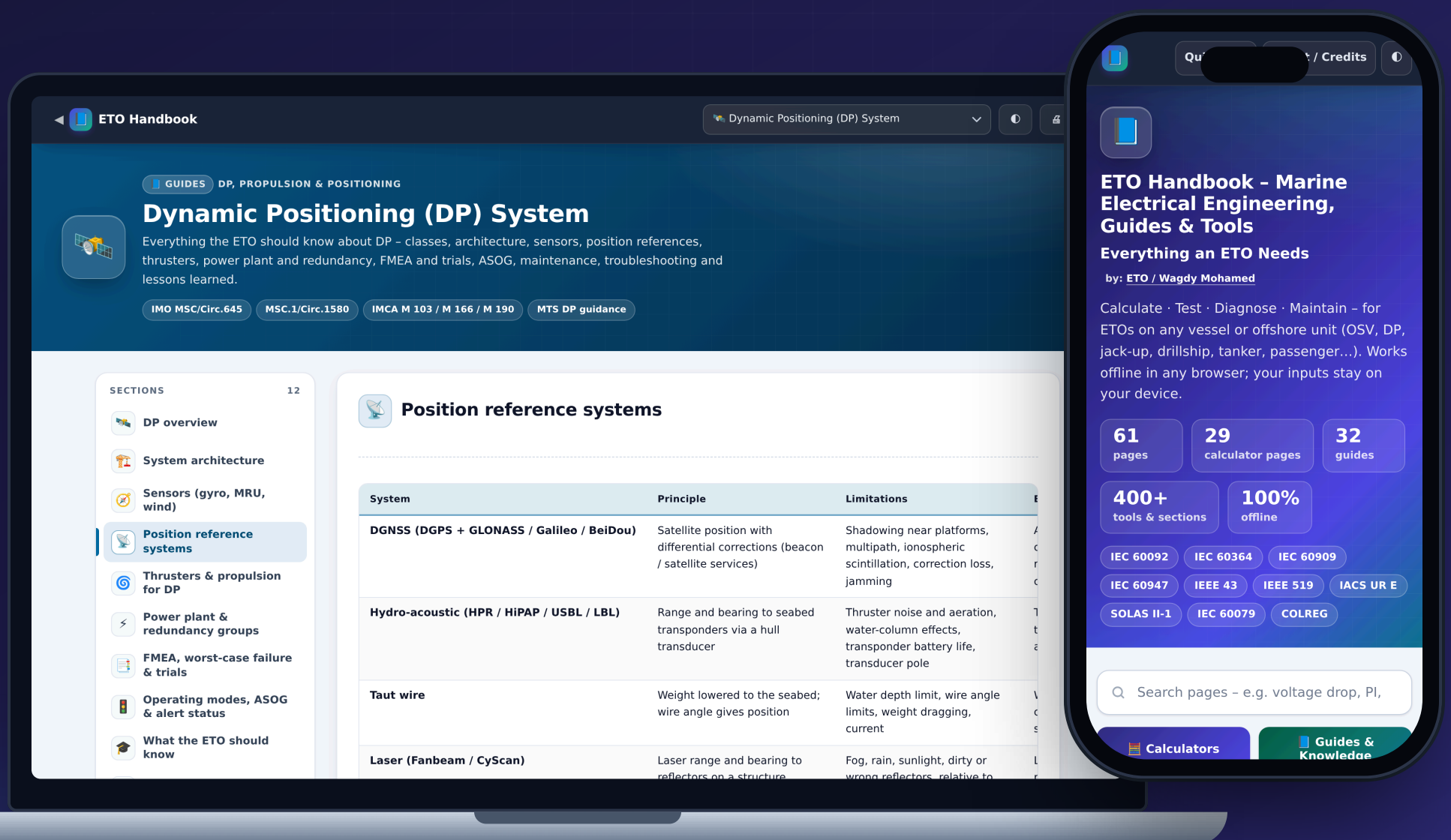




ETO HANDBOOK

Everything an ETO Needs

One free online handbook for marine & offshore Electro-Technical Officers.



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by: ETO / Wagdy Mohamed

29 calculator pages, 400+ calculations

Enter your nameplate data – get results, the formula used and a clear  /  /  verdict.

ETO Handbook

Short-circuit & Protection

Print

SECTIONS12

Prospective short-circuit current

Breaker breaking-capacity check

Breaker rating selection

Trip unit settings (LSI)

Earth-fault loop (TN) disconnection

RCD / earth leakage (TT)

Selectivity / discrimination (quick check)

Fuse rating

Motor protection settings

Protection coordination & time-current chart

IDMT relay (IEC 60255 curves)

CT burden & accuracy-limit factor

Transformer uk 4-6%; generator X'd 12-20%

Transformer 5-10; generator 15-30

Voltage

440V

Cable length from source

40m

Cable size (per phase)

240mm²

Parallel cables

2

Cable reactance

0.08mΩ/m

Running motor FLC total

300A

Motors running at the time of fault

Reset to defaults

Copy results

RESULTS

At source terminals

32.80kA

At fault point (source+cable)

26.45kA

Motor contribution

1.20kA

Total Isc (rms)

27.65kA

X/R at point

3.9

Peak current ip

57.7kA

Motor contribution ≈ 4 × FLC of running motors (LV). The max fault (all DGs + transformers) governs breaker Icu; the min fault (one DG, arcing) governs trip sensitivity.

Crane

Earthing & Insulation

About / Credits

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 Power & distribution

 Machines & drives

 Protection & PQ

 Power plant & DP

 Instrumentation

 Maths & units

2 / 6



32 system guides written for the ETO

Overview · structure · what the ETO should know · maintenance · troubleshooting · lessons.

1 Power, electrical & automation

2 DP, propulsion & positioning

3 Jack-up & offshore systems

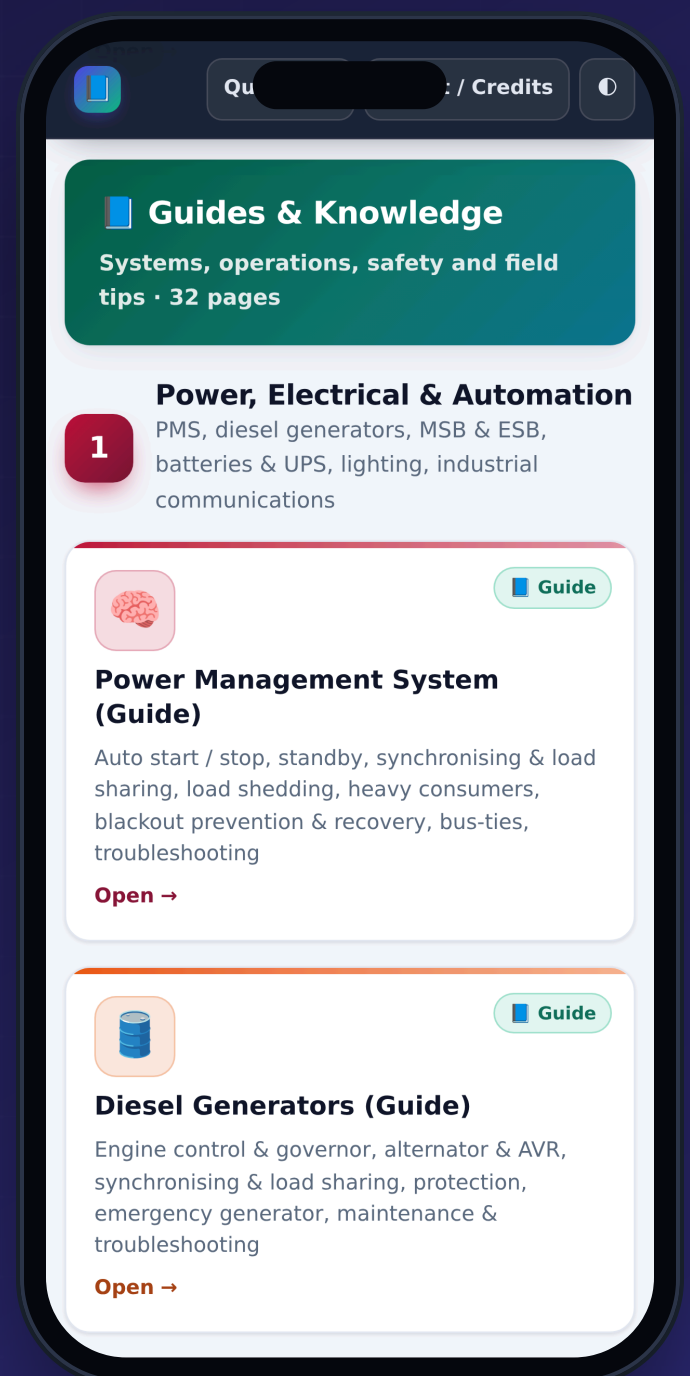
4 Auxiliary & marine systems

5 Safety systems & Ex areas

6 Operations & maintenance

7 Tips & tricks for ETO

DP · thrusters · jacking · jetting · raw water · PMS · diesel generators · MSB & ESB · UPS · ESD · fire & gas · cranes · pumps · valves · air · fuel & purifiers · water maker · sewage · BWMS · chillers...





Test yourself – 192 quiz questions

Every system guide ends with 8 questions. Tap an answer, see why it is right, get your score.

ETO Handbook

Diesel Generators (Guide)

Print

SECTIONS 11

Overview

Genset structure

Engine control, governor & safeties

Alternator, excitation & AVR

Synchronising, load sharing & PMS

Protection & alarms

Emergency generator

What the ETO should know

Maintenance & testing

Troubleshooting

Safety & lessons

Quick quiz

Quick quiz

Check how well you understand this system – 8 questions. Tap an answer to see whether it is right and why.

1 What is the typical overspeed shutdown setting for a diesel generator?

A About 103-105 % rated speed

B About 125-130 % rated speed

C About 110-115 % rated speed

D About 95-100 % rated speed

Correct - The guide lists overspeed shutdown at typically 110-115 % rated speed.

2 A self-excited generator runs at rated speed but builds no voltage after a long lay-up. What is a likely cause?

A Loss of residual magnetism

B Governor droop set too high

C Reverse power relay tripped

D Space heater left switched on

Not quite – the answer is A. A self-excited alternator relies on residual magnetism; field flashing per maker restores it.

3 In a brushless alternator, what feeds DC to the main rotor field?

A Slip rings and carbon brushes from the AVR

B The PMG mounted on the stator frame

C The varistor across the main stator

D The rotating diode bridge rectifying exciter output

✓ Instant feedback

💡 Explanations

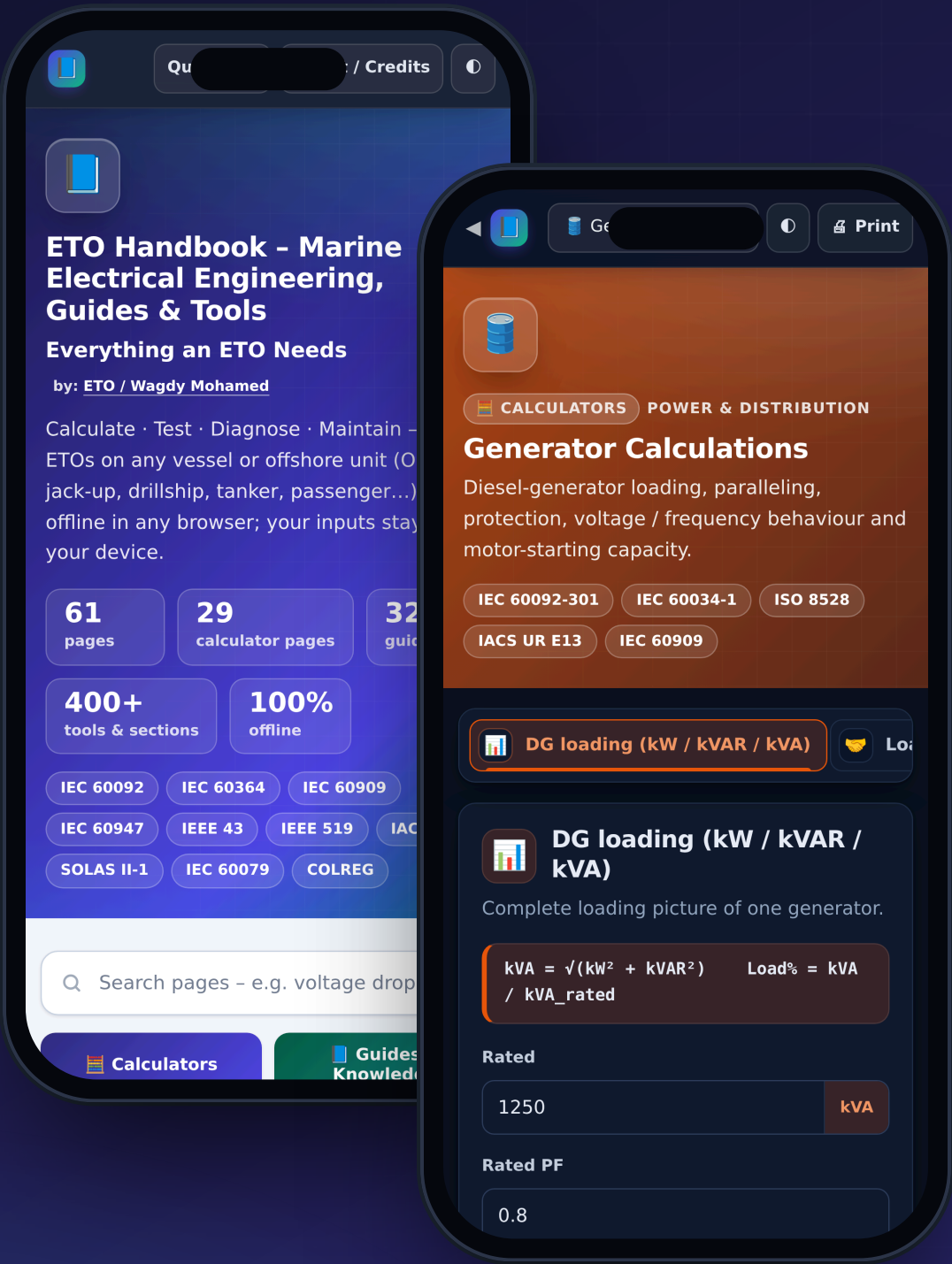
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